

Title

A research article title should be clear, concise, specific, informative, and easy to understand.

Key principles

Clear — communicates the research topic without ambiguity.

Concise — avoids unnecessary words.

Specific — identifies the main problem, method, subject, or contribution.

Informative — gives readers a good idea of what the paper contains.

Easy to understand — avoids unnecessary jargon, abbreviations, and complicated wording.

Example:

✗ *An Investigation into the Application of Various Advanced Deep Learning-Based Computational Techniques for the Purpose of Automated Detection of Vegetable Quality*

✓ *Deep Learning for Automated Vegetable Quality Detection*

Abstract

1. **Initialization** — *What is the research area?*

- Identify the broad research domain, topic, and current trend.
- Example: Machine Learning → Deep Learning → Image Classification

2. **Research Gap** — *What is missing?*

- Identify limitations, unanswered questions, weaknesses, or unexplored areas in existing research.
- What has not been adequately addressed by previous studies?
- Clearly establish the need for your study.

3. **Research Objective** — *What do you want to do?*

- *Clearly* define what the research intends to investigate, develop, improve, compare, or discover.
- What specific problem are you trying to solve?

4. **Proposed Approach** — *How will you do it?*

- Describe the methodology, models, algorithms, experimental design, dataset, and evaluation strategy.
- What approach will you use to address the research problem?

Abstract

5. **Justification** — *Why this Method and Dataset?*

- Explain why the proposed method, model, dataset, or experimental design is appropriate and necessary.
- Why should this approach be used instead of existing alternatives?

6. **Expected/Research Outcome** — *What will the research contribute?*

- Identify the expected findings, performance improvement, dataset, model, framework, insights, or other research contributions.
- What new knowledge, resource, method, or evidence will this research provide?

Abstract

A concise research journey:

➤ Research Area → Research Gap → Research Objective → Proposed Method → Justification → Experiment → Outcome → Contribution

OR

➤ Research Context → Gap Identification → Objective Formulation → Methodology Design → Justification → Experimental Evaluation → Research Contribution

1. What is the context?
2. What is the gap?
3. What do you want to do?
4. How will you do it?
5. Why is your approach necessary?
6. What did you achieve?

A simple formula: Context + Gap + Objective + Method + Rationale + Outcome = Research Abstract

Keywords

A simple approach is to identify **4–8 important concepts** from your research.

1. Identify the research topic

What is the paper mainly about?

Example:

Deep Learning for Automated Vegetable Quality Detection

Possible keyword: Deep learning, Vegetable quality

2. Identify the research method

What methods, algorithms, or technologies are used?

For example: Convolutional neural networks (CNN), Transfer learning, Object detection,
Computer vision

3. Identify the application/domain

Where is the research applied?

For example: Agricultural image analysis, Food quality assessment, Smart agriculture

Keywords

4. Identify the specific research problem

What problem does the research solve?

For example: Defect detection, Disease detection, Quality classification

5. Use standard technical terminology

Prefer terms that researchers actually use in the literature.

For example: ✗ *Computer method for finding bad vegetables*
✓ *Vegetable defect detection*

6. Avoid unnecessarily broad keywords

✗ Machine Learning ✗ Artificial Intelligence ✗ Computer Science

These are often too broad unless they are genuinely central to the paper.

Instead: ✓ Deep learning ✓ Object detection ✓ Vegetable defect detection ✓ Transfer learning

Keywords

A useful keyword formula

Keywords are selected from these categories:

[Research Domain] + [Problem] + [Method] + [Application] + [Specific Technique]

Paper title: Real-Time Detection of Fresh and Defective Vegetables Using Deep Learning

Keywords:

Deep learning; vegetable defect detection; object detection; computer vision; transfer learning; agricultural image analysis.

Keywords should represent the core topic, research problem, methodology, application domain, and distinctive contribution of the study.

Introduction Structure(1st para)

1. What is currently happening/ongoing? — Current Status

- What is the current situation in this research area?
- What are the existing methods, technologies, or approaches?
- What are the recent developments or trends?
- Why is this topic important?

2. What is missing? — Research Gap / Challenges / Research Questions

- What limitations exist in current research?
- What challenges remain unresolved?
- What problems are not adequately addressed?
- What research questions arise from these gaps?

3. What do you want to do and how? — Research Objective & Approach

- What is the main objective of your research?
- What do you propose or investigate?
- How will you address the identified gap?
- What method, model, framework, or approach will you use?

Introduction Structure(1st para)

4. What are the scopes of your Proposed Method and Dataset? — Scope & Applicability

- What can your proposed method do?
- Where can it be applied?
- What does the dataset contain?
- Who or what can use the dataset?
- What types of research, experiments, or applications can be supported by your method and dataset?
- What are the boundaries or limitations of the proposed scope?

A Simple Formula

- Current State → Research Gap/Challenges → Objective & Approach → Scope & Applicability

Or, more simply:

- What is happening? → What is missing? → What will you do and how? → What is the Scope ?
→ Where can it be used?

Introduction Structure(2nd para)

Merged Literature Review:

Paragraph 2 — Existing Research + Research Gap

- What have previous researchers done?
- What methods, models, datasets, or approaches have been used?
- What are the strengths and limitations of existing studies?
- What are challenges or unresolved issues remain?
- Based on the literature, what is the specific research gap?

Introduction Structure(3rd para)

Paragraph 3 — Research Objective + Proposed Approach

- What do you want to do?
- What is your research objective?
- How will you address the identified gap?
- What method/model/framework do you propose?
- What dataset will you use or develop?

Introduction Structure

Overall Flow

P1: Current State/ongoing



P2: Existing Research + Critical Literature Review + Research Gap



P3: What were not done/solve/resolve /mitigate and why in literature?

What You Want to Do + How You Will Do It?



P4: Scope of Proposed Method + Dataset + Contribution

Types of article

1. Method based
2. Dataset based
3. Method +dataset based

Organization of the article(Method based)

1. Introduction
2. Conventional method/methods
3. Proposed method
4. Experiments
 - 4.1 Experimental setup
 - 4.2 Corpus/Dataset

5. Results

6. Conclusion

Author contribution

Declaration

- Funding
- Data Availability Statement
- Competing Interests

Acknowledgements

Reference

Organization of the article(Corpus based)

1. Introduction

2. Dataset/Corpus

2.1 Data collection

2.2 -----

-----Details about dataset with several subsections-----

3. Conventional method(s) (if applicable)

4. Proposed method

5. Experiments

5.1 Experimental setup

6. Results

7. Conclusion

Author contribution

Declaration

- Funding
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Acknowledgements

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Organization of the article(Corpus+method-based)

1. Introduction
 2. Dataset/Corpus
 - 2.1 Data collection
 - 2.2 -----
-----Details about dataset with several subsections-----
 3. Conventional method/methods
 4. Proposed method
 5. Experiments
 - 5.1 Experimental setup
 6. Results
 7. Conclusion
- Author contribution
- Declaration
- Funding
 - Data Availability Statement
 - Competing Interests
- Acknowledgements
- Reference

2. Corpus/Dataset

For Dataset-based+Method-based Paper:

For a **dataset-based research paper**, the **Corpus/Dataset** should normally be introduced immediately after the **Introduction**, because the dataset itself is the primary contribution of the paper.

2. Corpus/Dataset

2.1 Dataset Overview

- Purpose and scope
- Dataset characteristics
- Target application/domain

2.2 Data Collection

- Data sources
- Collection procedure
- Collection period
- Inclusion/exclusion criteria

2. Corpus/Dataset

For Dataset-based+Method-based Paper:

2.3 Data Preparation

- Cleaning
- Preprocessing
- Filtering
- Formatting

2.4 Annotation/Labeling

- Annotation methodology
- Label/class definitions
- Annotation guidelines
- Annotators
- Quality-control procedure
- Inter-annotator agreement, if applicable

2. Corpus/Dataset

For Dataset-based+Method-based Paper:

2.5 Dataset Characteristics

- Number of instances
- Classes/categories
- Feature description
- Class distribution
- Statistical characteristics

2.6 Dataset Organization and Format

- File/folder structure
- Data format
- Metadata
- Naming conventions
- Train/validation/test organization, if provided

2. Corpus/Dataset

For Dataset-based+Method-based Paper:

2.7 Dataset Validation and Quality Assessment

- Quality checking
- Duplicate detection
- Missing/inconsistent data
- Label verification
- Data quality statistics

2.8 Ethical, Privacy, and Licensing Considerations

- Consent, where applicable
- Privacy/anonymization
- Ethical approval, if applicable
- Data license and usage restrictions

2 or 3. Conventional method/methods_(method-based sect.2,datasetbased sect.3)

The **Conventional Method(s)** section presents the baseline or existing method(s) from which the proposed method is derived or developed. The purpose is to establish a clear **technical foundation** for understanding the proposed approach.

Baseline Method

- Introduce the conventional or established method.
- Explain why it is selected as the baseline.
- Describe its main components and assumptions.

Working Principle

- Explain how the conventional method operates.
- Present the processing steps or workflow.
- Demonstrate the relationship between its input, intermediate processes, and output.

Technical Formulation

- Describe the method mathematically where appropriate.
- Include relevant equations, mathematical formulations, parameters, and objective functions.
- Define important variables and terms clearly.

2 or 3. Conventional method/methods

Algorithm / Procedure

- Present the method as an algorithm or step-by-step procedure when appropriate.
- Explain the key computational operations and decision processes.

Main Contribution / Characteristics of the Baseline

- Highlight the key technical contribution or distinctive characteristics of the conventional method.
- Explain what makes the baseline method useful or widely applicable.

Limitations and Research Gap

- Identify the specific limitations, weaknesses, or unresolved issues of the conventional method.
- Explain how these limitations motivate the development of the proposed method.
- End the section with the specific gap that the proposed method intends to address.

2 or 3. Conventional method/methods

Recommended Flow

Conventional Method → Working Principle → Mathematical Formulation → Algorithm → Key Characteristics of Baseline → Limitations → Research Gap → Proposed Method

Important Distinction

The Conventional Method section should not simply repeat the Literature Review. The Literature Review answers:

“What have previous researchers done?”

Whereas the Conventional Method section answers:

“How does the baseline method technically work, and what limitation motivates the proposed method?”

This makes the transition to the Proposed Method much clearer:

Baseline Method → Identified Limitation → Proposed Modification/Extension → Proposed Method

3. Proposed method

The **Proposed Method** section can follow a parallel structure with the Conventional Method, but it should focus on the **novelty, modification, and technical contribution** of your work.

The Proposed Method section describes the newly developed approach introduced to address the research gap identified in the Conventional Method. It should clearly explain **how the proposed method works, how it is technically formulated, how it is implemented, and what its main contribution is.**

3.1 Working Principle

- Explain the overall working mechanism of the proposed method.
- Describe the input, processing steps, and output.
- Explain how the proposed method differs from the conventional/baseline method.
- Use a **flowchart, block diagram, or conceptual framework** where appropriate.

3.2 Technical Formulation

- Present the mathematical formulation of the proposed method.
- Define equations, parameters, variables, constraints, objective functions, or other technical components.
- Clearly show any **modification, extension, or newly introduced component** compared with the conventional method.

3. Proposed method

3.3 Algorithm / Procedure

- Present the proposed method as a step-by-step algorithm or procedure.
- Clearly describe the computational workflow.
- Identify the major inputs, operations, decision processes, and outputs.
- Provide sufficient detail so that the method can be **understood and reproduced**.

3.4 Main Contribution / Characteristics of the Proposed Method

- Clearly identify what is new or different in the proposed method.
- Explain how it addresses the limitation or gap of the conventional method.
- Highlight the key technical characteristics and expected advantages.
- Distinguish the **novel contribution** from components adopted from existing methods.

3. Proposed method

Overall Methodology Flow

Conventional Method

Working Principle → Technical Formulation → Algorithm → Limitations → Research Gap



Proposed Method

Working Principle → Technical Formulation → Algorithm/Procedure → Main Contribution/Characteristics

This creates a very clear research narrative:

Existing Method → Limitation → Research Gap → Proposed Solution → Technical Novelty

A useful principle: *In the Proposed Method section, the reader should be able to answer three questions clearly:*

- 1.What exactly is proposed?
- 2.How does it work technically?
- 3.What is new compared with the conventional method?

4. Experiments

For Method-based Paper:

4.1 Experimental Setup and Conditions

- Hardware configuration
- Software/platform
- Programming environment
- Experimental parameters
- Simulation or implementation conditions
- Reproducibility settings

4.2 Dataset Introduction (This section is only for method based. For dataset-base it is avoided)

- Dataset source
- Dataset size
- Number of classes/features
- Data characteristics
- Data distribution
- Training/validation/testing split
- Preprocessing, if applicable

4. Experiments

For Method-based Paper:

4.3 Experimental Procedure

- Overall experimental workflow
- Data preparation
- Application of the conventional/baseline method
- Application of the proposed method
- Training/testing procedure
- Repeated experiments or cross-validation
- Experimental scenarios

4.4 Parameter Configuration

- Model/method parameters
- Hyperparameters
- Fixed parameters
- Optimization settings
- Parameter selection/tuning strategy

4. Experiments

For Method-based Paper:

4.5 Evaluation Metrics

- Accuracy, Precision, Recall, F1-score, etc., where applicable
- Error/performance measures
- Mathematical formulation of metrics
- Statistical measures, if applicable

Recommended structure

4.1 Experimental Setup and Conditions



4.2 Dataset Introduction



4.3 Experimental Procedure



4.4 Parameter Configuration



4.5 Evaluation Metrics

5. Results

Results should be presented and analyzed based on the defined evaluation metrics, because the metrics provide the basis for interpreting and comparing the experimental outcomes.

5.1 Results Based on Evaluation Metrics

- Accuracy-based Results
- Precision-based Results
- Recall-based Results
- F1-score-based Results
- And so on

5.2 Comparative Results

- Conventional/baseline vs. proposed method
- Metric-wise comparison
- Improvement/reduction in performance measures

5. Results

5.3 Ablation Study (if applicable)

An ablation study in machine learning is an experiment where you remove, disable, or replace specific components (like layers, features, or modules) of a model to measure how much they contribute to its overall performance.

5.4 Statistical Analysis

- Confidence Interval
- Effect size
- t/p value

5.5 Discussion of Experimental/Key Findings

- Interpretation of the metric-wise results
- Why the proposed method performs differently
- Strengths and limitations observed from the experiments

Recommended structure

Metric-wise Analysis → Comparison → Ablation → Statistical Validation → Findings

6. Conclusion

A **Conclusion** section should briefly summarize the research, the main contribution, the key findings, and possible future work.

Conclusion contains:

- Research objective/problem – **What was addressed?**
- Proposed work/contribution – **What was developed or introduced?**
- Key findings – **What were the major experimental outcomes?**
- Research significance – **What do the findings demonstrate?**
- Future work – **What can be improved or explored next?**

Recommended structure

Conclusion = Objective → Contribution → Key Findings → Significance → Future Work

Important: Avoid introducing new results, new methods, or new references in the Conclusion.

Author Contributions

“Who did what in this research?”

Authors may contribute to:

Conceptualization – Research idea, objectives, and overall research goals

Methodology – Development/design of the methodology or models

Software – Programming, implementation, and code development

Validation – Verification and reproducibility of results

Formal Analysis – Statistical, mathematical, or computational analysis

Investigation – Conducting experiments and data/evidence collection

Resources – Providing materials, datasets, computing resources, etc.

Author Contributions

“Who did what in this research?”

Authors may contribute to:

Data Curation – Managing, cleaning, annotating, and maintaining research data

Writing – Original Draft – Preparing the initial manuscript

Writing – Review & Editing – Reviewing, revising, and editing the manuscript

Visualization – Preparing figures, graphs, and other visualizations

Supervision – Research supervision and mentorship

Project Administration – Coordination and management of the research project

Funding Acquisition – Obtaining financial support for the research

Declaration

Declaration part includes

- Funding
- Conflict of interest
- Data availability
- Code availability
- Ethical approval statement

Acknowledges

The **Acknowledgements** section recognizes people or organizations that supported the research but do not qualify for authorship.

Acknowledgements may include:

Technical assistance — laboratory, programming, equipment, or technical support

Research assistance — help with data collection, annotation, experiments, or other research activities

Administrative support — institutional or departmental assistance

Language/editing support — proofreading or language improvement, where appropriate

Data/resource providers — individuals or organizations that provided datasets, materials, or other resources

Institutional support — departments, laboratories, universities, or research centers

Funding support — if the journal does not require funding information in a separate section

Special assistance — individuals who provided valuable advice or feedback but did not meet authorship criteria

References

References

- ❖ Use recent and relevant literature, including contemporary journal articles, conference papers, symposium proceedings, book chapters, and books, where appropriate.
- ❖ Ensure that the references are relevant to the research topic and sufficiently up to date.
- ❖ Present all references in a consistent and uniform citation style, such as IEEE, throughout the manuscript.
- ❖ Avoid unnecessary URLs/link-based references. When a URL is required, provide the access date according to the selected reference style or journal guidelines.
- ❖ Include bibliographic information such as author(s), article/chapter title, journal/conference/book title, volume, issue, page range, publication date/year, publisher, DOI, etc., as applicable.

References

References

- ❖ Follow the reference-formatting requirements of the target journal/conference. The journal's guidelines take precedence over a generic reference format.
- ❖ Ensure that every in-text citation appears in the reference list and every listed reference is cited in the manuscript.
- ❖ Verify the accuracy of author names, article titles, publication details, DOI, volume, issue, and page numbers before submission.

One important correction:

“Clearly mention Vol. number, Issue, PP., date, publisher” is not universally applicable.

For example, an IEEE journal reference may contain volume, issue, pages, and year, while a conference paper, book, book chapter, or article with an article number may have a different structure.